V4 visual and table comparison

Source baseline: PR #1 commit 0aa1c9b3. Revised: current V4 evidence reintegration.

Pages 2-3: Table 3 and Figure 2, original p7 then revised p7.

Page 4: Table 3 prose alternative, real manuscript p7 (not selected).

Pages 5-6: Figure 3 B vs C and Table 5, original p9 then revised p9.

Pages 7-8: Table 7, original p11 then revised p11.

The PDF pages are complete manuscript pages, not isolated cropped snippets.

Candidate C keeps Tire at camera004 and adds all ten frozen Bake10 effects.

No inferential result or statistical cohort was reselected for the graphic.

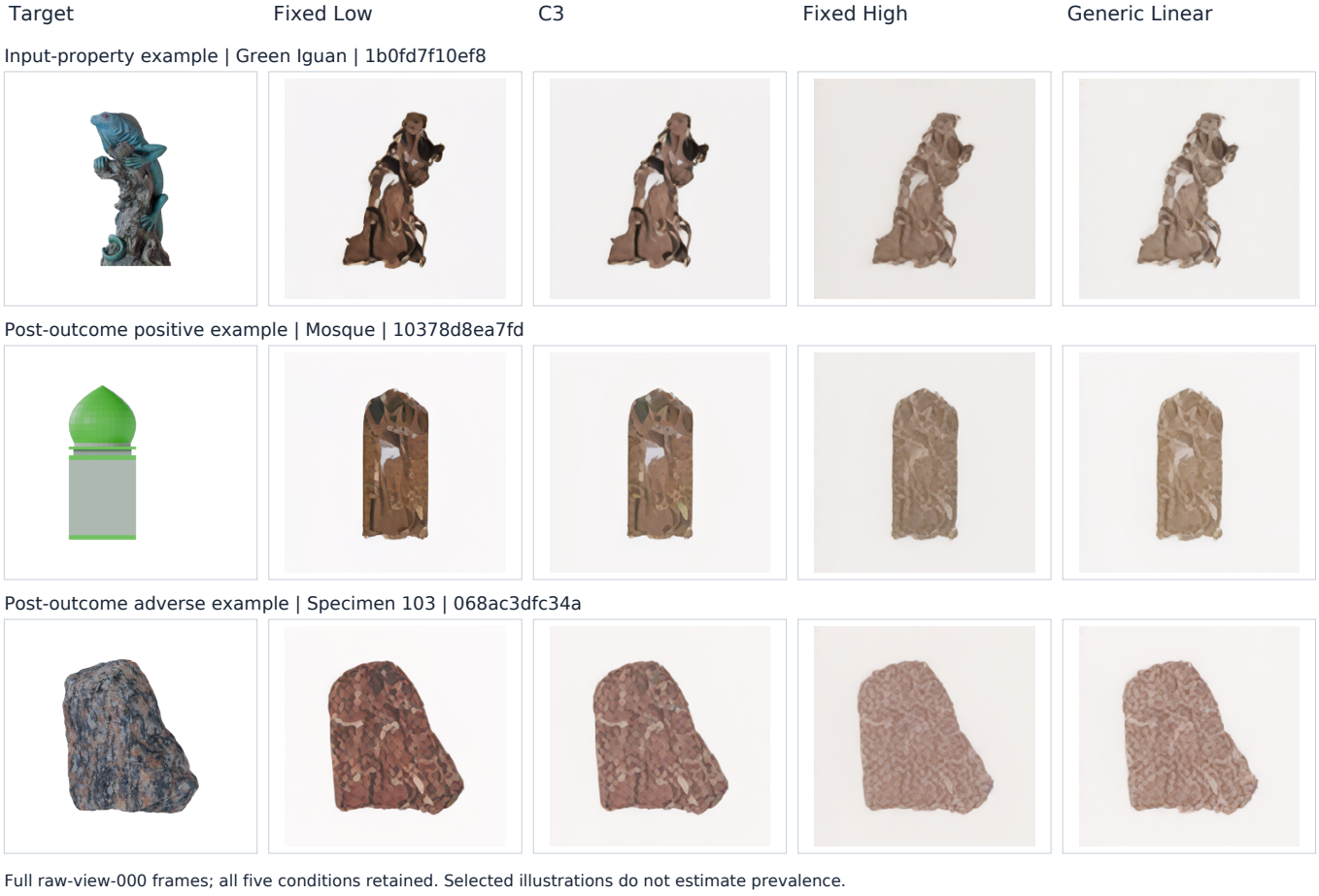


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

Table 3. Selected paired color-diagnostic contrasts.

Contrast	Mean CIEDE2000 difference [95% CI]	Objects
GFL – No Adapter	−19.654 [−26.370, −13.248]	24/26 lower
LLH – GFL	+3.583 [+0.462, +6.712]	–

The 26 objects are failure-enriched, not a prevalence sample. Negative differences favor the first condition.

a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation for that example, without identifying the complete cause.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss from it. Variation statistics alone cannot establish fidelity.

4.4. Actual Baked Appearance: Local Benefit and External Limits

We selected ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle count and camera-mask alignment. Only ten of the 300

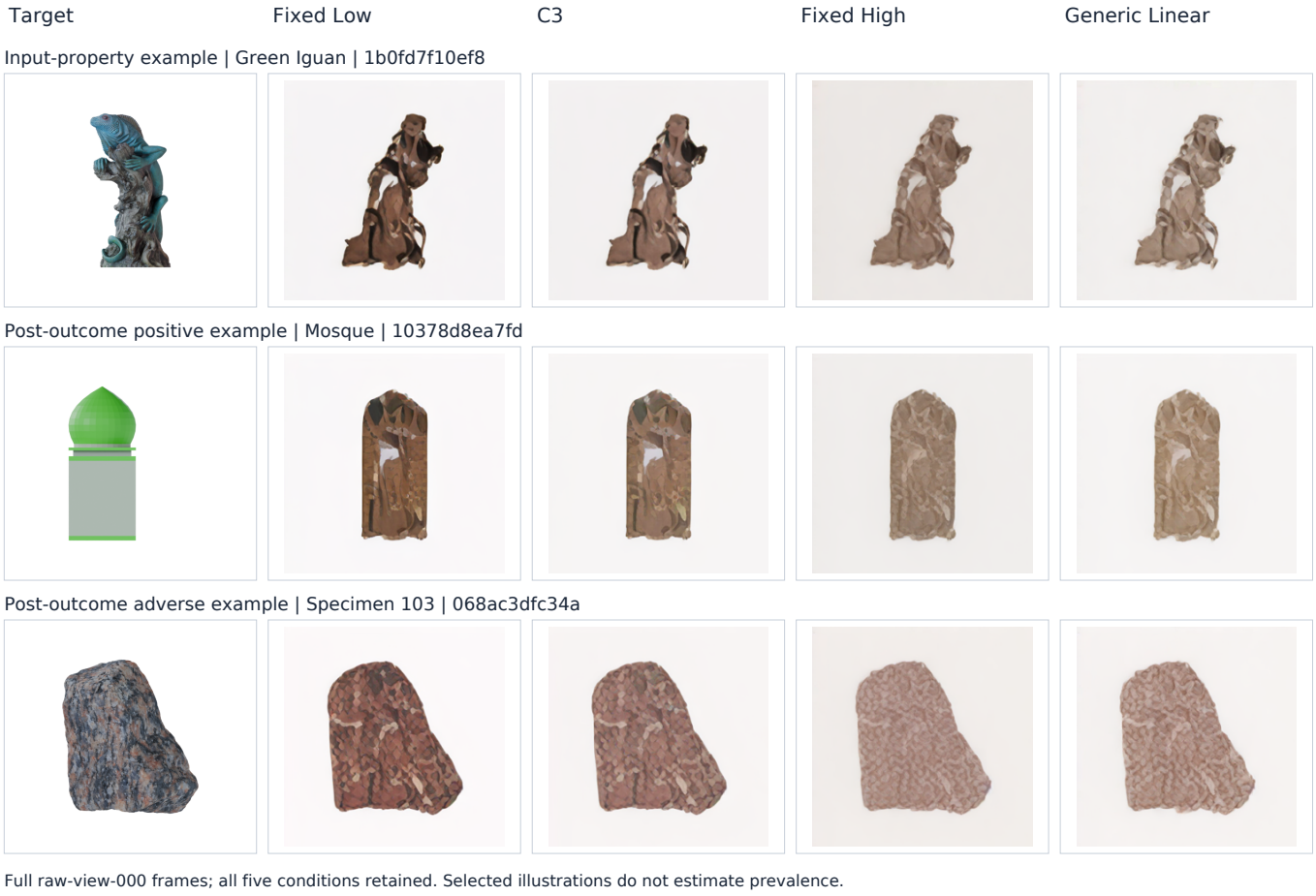


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Table 3. Selected paired color-diagnostic contrasts.

Contrast	ΔE_{00}	95% CI	Favorable
GFL – No Adapter	−19.654	[−26.370, −13.248]	24/26
LLH – GFL	+3.583	[+0.462, +6.712]	–

Failure-enriched $N = 26$; negative differences favor the first condition. Favorable is an object-direction count, not prevalence.

seed 61008), and no confirmatory multiplicity family was defined.

GFL lowered reference-relative CIEDE2000 in 24/26 objects, while visible pink or purple shifts remained in two legacy examples under No Adapter, GFL, and LLH. Their mapping to the original submitted panels cannot be recovered; the Fresh C example in Figure 4 is a different object. The shifts are present in decoded RGB before UV projection or baking. In one case, a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation for that example, without identifying the complete cause.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

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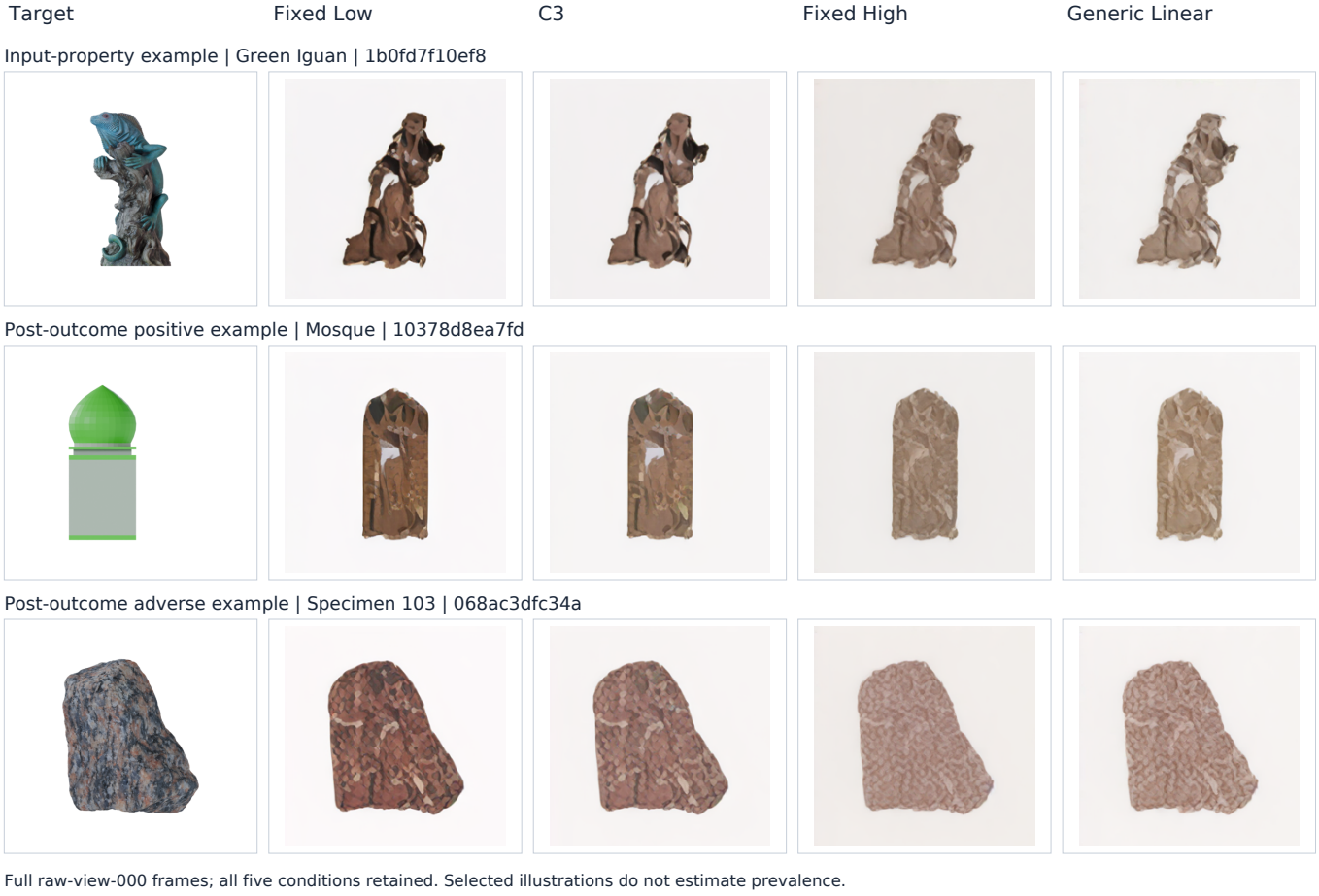


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

seed 61008), and no confirmatory multiplicity family was defined.

GFL minus No Adapter reduces mean CIEDE2000 by −19.654 (pointwise 95% paired-object interval [−26.370, −13.248]), with 24/26 objects lower; LLH minus GFL increases it by +3.583 ([+0.462, +6.712]). These descriptive comparisons concern a failure-enriched cohort, have no confirmatory multiplicity family and do not establish material fidelity; the full endpoints remain in Supplementary Section S6.

GFL lowered reference-relative CIEDE2000 in 24/26 objects, while visible pink or purple shifts remained in two legacy examples under No Adapter, GFL, and LLH. Their mapping to the original submitted panels cannot be recovered; the Fresh C example in Figure 4 is a different object. The shifts are present in decoded RGB before UV projection or baking. In one case, a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation for that example, without identifying the complete cause.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss



Fig. 3. GeoTex Bake10: GT-256 self-bake, Fixed Mean and Late-high at the same held-out unseen camera 004. Tire is selected by the median foreground Laplacian variance of the seven cleared source inputs; this input-property example is not a population sample. The baptismal font is explicitly selected post-outcome for the largest existing eleven-view Late-high–Fixed Mean FG-LPIPS reduction among those seven. The respective reductions are -0.002791 **and** -0.007810**; the latter accompanies lower FG-PSNR. The displayed differences are subtle compared with the remaining reference-appearance mismatch. All PNGs are 512×512, with one shared crop per row and common mesh/UVs within each object. GT-256 is a matched reference self-bake. Row scores average eleven views, not only camera 004. Credits and selection records are in Supplementary Sections S5/S9 and the V4 source manifest.**

$[0.440, 0.733]$). A post hoc sensitivity restricted to the seven original scale/time controls gives 0.396 ($[0.175, 0.582]$), with the minimum-mean LPIPS condition changing on 11/20 objects. These are positively associated rankings with changing point-estimate winners, not general rank reversal or statistically resolved winner changes. The GT self-bake has mean LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

Incoming-view consistency. The bake also measures cross-camera color variance at UV texels receiving multiple source views. A constant-black control has zero variance despite its incorrect appearance, showing that this diagnostic measures incoming-view disagreement rather than fidelity. Late-high has a lower mean disagreement than Fixed Mean (-0.002448), while Fixed High has still lower disagreement and higher baked LPIPS. Resampling and overlapping UVs also affect this measure; it does not validate seams or PBR. Full controls and contrasts are in Supplementary Section S5.

Native controls and protocol sensitivity. Subsequently added Native residual and No Adapter outputs on the same ten GeoTex objects are exploratory. They retain matched geometry and GT coverage, and the repeated Constant 1.0 (capped) outputs are bitwise identical. Their apparent PSNR/LPIPS gains for the schedules do not separate under the fifteen-test exploratory family. The complete data are in Supplementary Section S5.

A supporting fifty-object comparison uses a newly trained frozen GeoTex checkpoint with 512-pixel supervision, actual positive/negative reference conditions and CFG 2. Late-high minus Fixed Mean has nonreference PSNR -1.184 dB (interval $[-1.587, -0.849]$) and LPIPS $+0.00266$ ($[-0.00003, +0.00640]$; adjusted $p = 0.080$). C3 has no corrected advantage over Fixed Mean or Fixed High. This previously studied cohort and changed checkpoint/protocol supply a sensitivity boundary, not an independent backbone confirmation. Training and full-v29 strong-baseline results remain in Supplementary Section S8.

4.5. Allocation Responses in Other Conditioning Interfaces

MV-Adapter and MVDiffusion supply different conditioning operators and scale semantics from GeoTex. We evaluate each within its native target/camera contract and do not pool their scores or interpret identical scale labels as common residual doses.

New native MV-Adapter temporal comparison. We selected fifty Fresh C inputs not previously evaluated through MV-Adapter, using input availability and native material/geometry

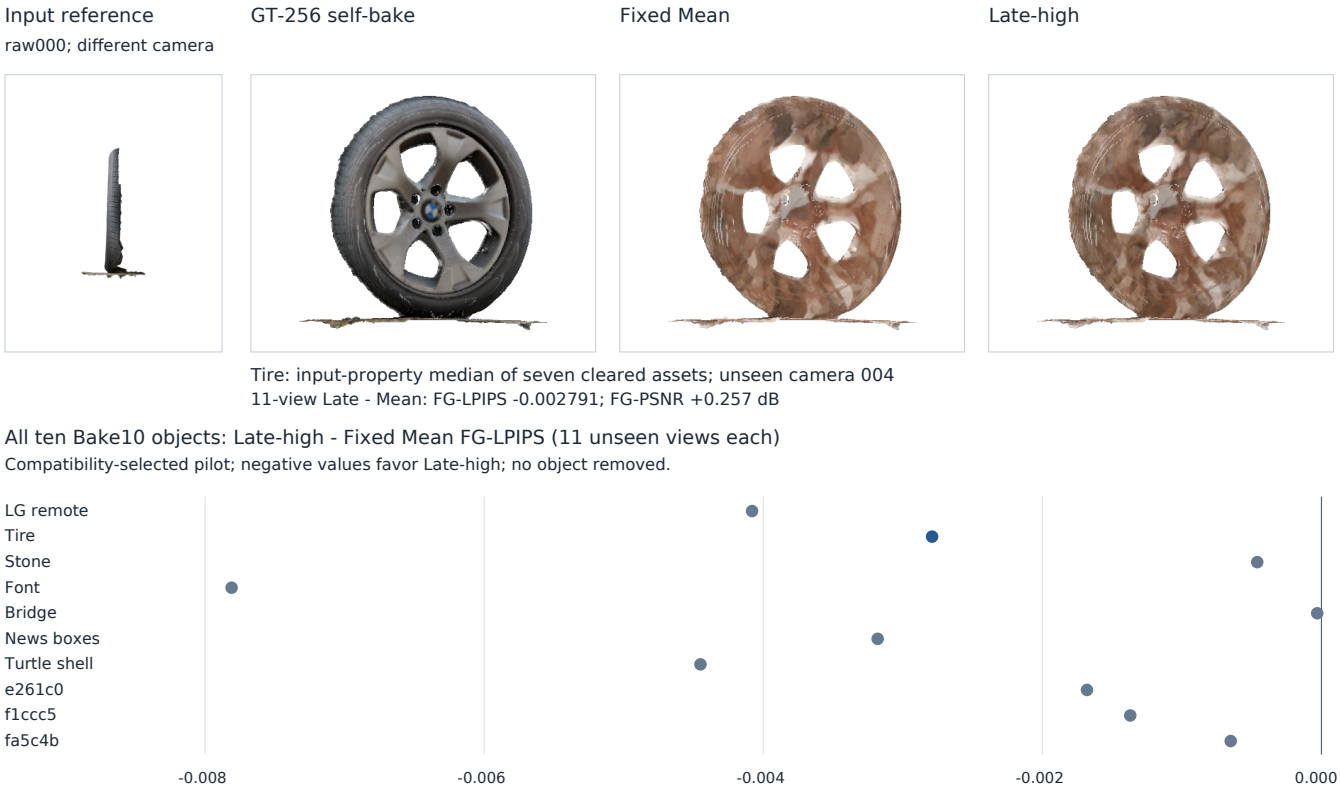


Fig. 3. GeoTex Bake10 asset appearance and object-level allocation effects. The small input inset is raw view 000 and is not pose-matched to the three exported-GLB renders at held-out camera 004. Tire is the median source-texture-complexity input among seven display-cleared assets, not a population-representative object. GT-256, Fixed Mean and Late-high use one shared crop and the same mesh/UVs; all source PNGs are 512×512, without method-specific color edits. The lower plot shows every object in the compatibility-selected ten-object pilot using existing eleven-view Late-high minus Fixed Mean FG-LPIPS means; negative values favor Late-high. It describes heterogeneous, small metric effects, not visible material fidelity or a new inferential test. GT-256 is a reference self-bake, not an upper bound. Full UID values and conditions are in Supplementary Section S5.

Table 5. Unseen-view FG-LPIPS primary contrasts for the two bake studies.

Study (N)	Δ FG-LPIPS	95% CI	p
GeoTex (10)	−0.002652	[−0.004144, −0.001367]	0.00195
GSO (20)	+0.000084	[−0.000103, +0.000286]	0.788

Late-high minus Fixed Mean. GeoTex: exact paired sign-flip. GSO: original-family Holm16-adjusted paired sign-flip. Separate input and calibration protocols are not pooled; non-detection is not equivalence.

LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

Incoming-view consistency. The bake also measures cross-camera color variance at UV texels receiving multiple source views. A constant-black control has zero variance despite its incorrect appearance, showing that this diagnostic measures incoming-view disagreement rather than fidelity. Late-high has a lower mean disagreement than Fixed Mean (−0.002448), while Fixed High has still lower disagreement and higher baked LPIPS. Resampling and overlapping UVs also affect this measure; it does not validate seams or PBR. Full controls and contrasts are in Supplementary Section S5.

Native controls and protocol sensitivity. Subsequently added Native residual and No Adapter outputs on the same ten GeoTex objects are exploratory. They retain matched geometry and GT coverage, and the repeated Constant 1.0 (capped) outputs are bitwise identical. Their apparent PSNR/LPIPS gains for the schedules do not separate under the fifteen-test exploratory family. The complete data are in Supplementary Section S5.

A supporting fifty-object comparison uses a newly trained frozen GeoTex checkpoint with 512-pixel supervision, actual positive/negative reference conditions and CFG 2. Late-high minus Fixed Mean has nonreference PSNR −1.184 dB (interval $[-1.587, -0.849]$) and LPIPS +0.00266 ($[-0.00003, +0.00640]$; adjusted $p = 0.080$). C3 has no corrected advantage over Fixed Mean or Fixed High. This previously studied cohort and changed checkpoint/protocol supply a sensitivity boundary, not an independent backbone confirmation. Training details and full-v29 comparisons are in Supplementary Section S8.

Native-pipeline quality boundary. The recorded full-v29 ControlNet pipeline is a stronger deployment comparator than geometry-disabled output from the new HF-based checkpoint. On the same fifty inputs and five nonreference cameras, the

Table 7. Specified FG-PSNR schedule primaries.

Study and contrast	N	FG-PSNR difference [95% CI], dB	p / test
GeoTex Fresh C: Middle-high − mean(Early-high, Late-high)	300	−0.033877 [−0.117170, +0.051422]	$p = 0.432378$; paired sign-flip, 20,000 draws
MV-Adapter50: Middle-high − Fixed Mean	50	+0.028986 [−0.019411, +0.080739]	$p = 0.280747$; paired Monte Carlo sign-flip, 100,000 draws
GSO100: Middle-high − Fixed Mean	100	−0.013660 [−0.030565, +0.003519]	$p = 0.122279$; paired Monte Carlo sign-flip, 100,000 draws

Each row is its study's specified primary. Intervals are pointwise 95% paired-object intervals; all estimates are FG-PSNR in dB. The GeoTex Fresh C row is Middle-high minus mean(Early-high, Late-high); a separate Middle-high minus Fixed Mean comparison does not replace it. The studies remain separate, and matching applied scalar sums does not establish equal realized residual dose.

mixed or tied directions. Its nonreference LPIPS averages change direction (−0.000389, −0.000269, +0.000398). For the GSO20 bake, only 1/20 objects has lower Late–Mean LPIPS at all three seeds; 14/20 have mixed or tied directions. Sign agreement has no practical-effect threshold. These conditional, object-level summaries do not estimate training randomness or revise the original primary tests; full seed intervals and deltas are in Supplementary Section S4.

The retrospective 75-object MVDiffusion panel scales correspondence-block outputs rather than the GeoTex residual operator. Its layer-wise L–LLH condition is worse than global fixed low in PSNR, structural agreement and reference-color error; LPIPS does not separate. This result and the earlier MV-Adapter panel remain in Supplementary Section S3 and delimit the scope of cross-interface transfer.

Projection-guidance candidates are retained as exploratory negative results in Supplementary Sections S4 and S8; they do not form a validated alternative method.

4.6. Historical Cohort Sensitivity and Prior Evaluation

The 276-slot retraining tests endpoint sensitivity on a previously exposed historical request list. Seventy-nine requested UIDs overlapped the earlier recorded training list and were removed, leaving 1,039 training UIDs with no requested-list overlap. During evaluation, an empty-input error at requested slot 46 triggered an inherited loader's random replacement whose actual UID was not recorded. The 1,104 rows therefore represent 276 requested slots under four paired conditions, not verified coverage of 276 distinct held-out objects. This panel is reported as a protocol/checkpoint sensitivity analysis; it does not recover the historical main checkpoint or reproduce its reported +0.96 dB result.

C3 minus Fixed High is +1.699 dB in full-frame PSNR (276/276 slots positive), but −1.314 dB in FG-PSNR (9/276 favorable). FG-LPIPS is higher by 0.01127, FG-SSIM lower by 0.05961, and Edge-SSIM lower by 0.02777. Excluding the known failed requested slot preserves the opposing PSNR means (+1.697 and −1.311 dB) and the LPIPS loss (+0.01127). White-background normalization also favors Fixed High. The raw full-frame gain therefore does not support reference-texture fidelity. C3 improves FG-PSNR over Fixed Low by 0.783 dB, while their descriptive FG-LPIPS interval includes zero. Slot-bootstrap diagnostics do not establish independent-object inference because the substituted UID is unknown. Fixed Mean, moved high windows and Generic Linear were not evaluated here, so this panel cannot test temporal allocation. The requested-slot condition means (Supplementary Table S36) and paired endpoint contrasts (Supplementary Table S35) remain in Section S7, with the unknown actual-UID substitution and technical, non-confirmatory scope stated alongside.

The original blinded 3AFC study on thirty objects and twenty-four participants reports C3's overall-quality share as 58.1% (two-level participant/object-bootstrap interval [50.1, 65.8]%). This preference applies to the original stimuli; it does not measure texture fidelity for the retrained checkpoint or unseen baked appearance. Counts and endpoint-specific results are retained in Supplementary Section S7.

Historical learned-control comparison. Previously reported Full Adaptive Correction (FAC) configurations all reduced FG-PSNR relative to TCAS in their stated controlled re-examination. The original training pool was reported as 1,706 objects, while a recovered configuration refers to 1,200; seed, checkpoint and sampled-run records are incomplete. This result is retained as historical context and does not establish a general comparison between learned and inference-time control. Source identity, recovered code and the limits of training replay are detailed in Supplementary Section S7.

5. Discussion and Limitations

Strength and timing answer different questions. A low–high–low curve can appear effective when the fixed comparator is weak, yet that contrast changes both amplitude and placement. Fresh C makes this distinction concrete: stronger competitive controls change the practical recommendation, and the preserved equal-budget primary does not support the earlier middle-window advantage. Separate pairwise results favor Fixed Mean and Late-high over C3 on several endpoints, while Late-high trades lower LPIPS and higher PSNR against lower FG-SSIM than Fixed Mean. These observations change the tested recommendation, but do not show that every temporal schedule is ineffective or that Late-high is universally best.

Matching scalar sums does not identify a residual-dose mechanism. The native shallow cap makes all three C3 requests identical after clipping in that group. A requested-profile comparison can therefore describe an intervention that was never applied as drawn. Reporting the implemented per-group profiles and sums removes that ambiguity. Matching those sums still leaves the adapter activation and hidden-state trajectory free to change. The matched-budget contrasts estimate the total effect of allocating the scalar control, including the resulting correction changes; they do not isolate timing at an equal realized

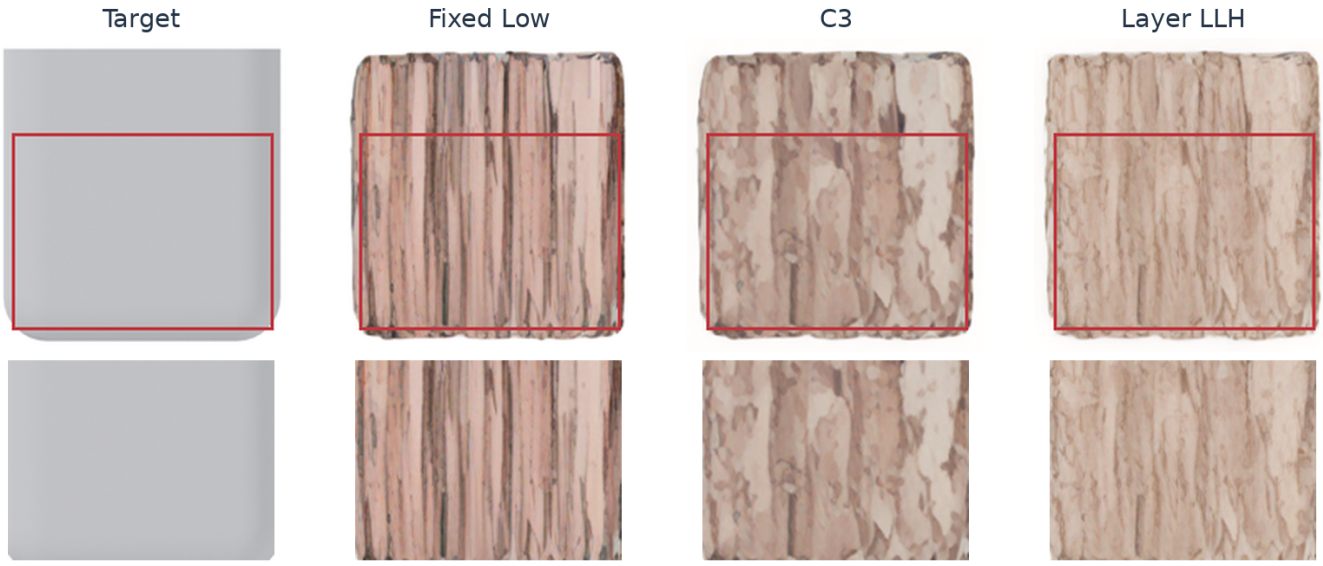


Fig. 4. Fresh C view 0, E5-5-2: Target, Fixed Low, C3 and layer LLH share a complete-object frame and an identical pixel crop. The generated brown pattern does not reproduce the gray target; changing its variation does not establish color fidelity. This previously viewed example came from an existing C3–GFL LPIPS stratum and was not selected for color. It does not reproduce the legacy purple-shift cases. Object credits are supplied in Supplementary Section S9.

Table 7. Specified FG-PSNR schedule primaries.

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Middle/Early/Late denote the corresponding high-window allocations. Each row is its study's specified primary; the GeoTex row is the original timing contrast, not Middle–Fixed Mean. Pointwise paired-object intervals; GeoTex uses 20,000 and the native-interface rows 100,000 paired sign-flip draws. Studies are separate; equal applied scalar sums do not imply equal realized residual dose.

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